

# Software Requirement Specification (SRS) for DBMS Project

## 1. Introduction

### 1.1 Purpose

The purpose of this project is to design and develop an **Intelligent Inventory Management System** that uses **data-driven forecasting** to predict product demand in the retail sector. The system uses **machine learning algorithms** on historical sales data to help businesses optimize inventory levels, reduce stockouts and overstocking, and enhance supply chain efficiency.

### 1.2 Scope

The system will:

- Maintain a centralized **database** for storing and managing inventory, customer, and transaction details.
- Use **machine learning** to forecast future product demand based on historical data.
- Integrate **relational (MySQL)** and **NoSQL (MongoDB)** databases for structured and unstructured data.
- Provide **real-time analytics dashboards** and **forecast summaries** using NLP for non-technical users.
- Generate **reports** on sales, demand trends, and product performance.

**Benefits to users:**

- Data-driven decision-making for stock control.
- Reduced waste and improved resource utilization.
- Enhanced business scalability and forecasting accuracy.

### 1.3 Definitions, Acronyms, and Abbreviations

| Term        | Abbreviations                | Definitions                                                                                                      |
|-------------|------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>DBMS</b> | Database Management System   | Software that lets you store, manage, and retrieve data efficiently.                                             |
| <b>ERD</b>  | Entity Relationship Diagram  | A diagram that shows how data tables (entities) are connected through relationships.                             |
| <b>SQL</b>  | Structured Query Language    | The standard language used to interact with databases—writing queries to insert, fetch, update, and delete data. |
| <b>CRUD</b> | Create, Read, Update, Delete | The four basic operations you can perform on data in any system.                                                 |

|              |                                       |                                                                                                                      |
|--------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>ETL</b>   | Extract, Transform, Load              | A data pipeline process where data is taken from a source, cleaned/modified, and loaded into another system.         |
| <b>ML</b>    | Machine Learning                      | A field where computers learn patterns from data and make predictions without being explicitly coded for every rule. |
| <b>NLP</b>   | Natural Language Processing           | Techniques that help computers understand, interpret, and generate human language.                                   |
| <b>LLM</b>   | Large Language Model                  | A very large neural network trained on massive text data to understand and generate human-like text (like GPT).      |
| <b>RDBMS</b> | Relational Database Management System | A DBMS that stores data in tables (rows and columns) with relationships between them—like MySQL, PostgreSQL, Oracle. |

## 2. Overall Description

### 2.1 Product Perspective

The system is a **standalone web-based application** that integrates both **RDBMS (MySQL)** and **NoSQL (MongoDB)** through a **Python-based ETL pipeline**. It is part of a broader **AI-driven business intelligence ecosystem** where MySQL serves as the system of record and MongoDB as the system of engagement.

### 2.2 Product Functions

Major functionalities include:

- **User Management:** Authentication for admin and business users.
- **Inventory Management:** Add, update, delete, and track stock details.
- **Sales Management:** Record and manage invoice and transaction data.
- **Demand Forecasting:** Predict future demand using machine learning models.
- **Data Integration:** ETL pipeline connecting MySQL and MongoDB.
- **Reports & Analytics:** Generate sales, forecast, and performance reports.
- **Backup & Recovery:** Regular data backup and restoration features.

### 2.3 User Characteristics

| User Type     | Description                               | Skill Level  |
|---------------|-------------------------------------------|--------------|
| Administrator | Manages system, users, and data integrity | Intermediate |

|                   |                                               |                       |
|-------------------|-----------------------------------------------|-----------------------|
| Inventory Manager | Oversees stock, sales, and demand forecasts   | Basic computer skills |
| Data Analyst      | Works with data insights, reports, and trends | Technical / Advanced  |

## 2.4 Constraints

- **Hardware:** Minimum 8GB RAM, 2GHz Processor.
- **Software:** MySQL 8.0, MongoDB 6.0, Python 3.10, Flask, Express, EJS .
- **Database Constraints:** Normalized up to 3NF.
- **Storage:** Dependent on business transaction volume.
- **Network:** Minimum stable bandwidth: 5 Mbps download / 2 Mbps upload; latency  $\leq$  150 ms; uptime  $\geq$  95%.

## 2.5 Assumptions and Dependencies

- Minimum requirement: 2–5 Mbps connection with latency  $\leq$  150 ms and uptime  $\geq$  95%.
- Input data (sales, customer info) is valid and complete.
- Machine learning models are retrained periodically.
- Python environment and required libraries (e.g., Pandas, Scikit-learn) are available.

## 3. Specific Requirements

This section defines the **functional and non-functional** requirements.

### 3.1 Functional Requirements

| Module             | Requirement ID | Description                                                                           |
|--------------------|----------------|---------------------------------------------------------------------------------------|
| User Management    | FR-1           | The system shall allow users to register, log in, and manage profiles.                |
| Inventory Module   | FR-2           | The system shall store and manage product details (StockCode, Description, Quantity). |
| Customer Module    | FR-3           | The system shall maintain customer details and purchase history.                      |
| Transaction Module | FR-4           | The system shall record invoice data, product sales, and pricing.                     |
| Forecasting Module | FR-5           | The system shall use ML models to predict future product demand.                      |
| Integration Module | FR-6           | The ETL pipeline shall synchronize MySQL and MongoDB data.                            |

|                            |      |                                                                      |
|----------------------------|------|----------------------------------------------------------------------|
| <b>Report Generation</b>   | FR-7 | The system shall generate sales and demand forecast reports.         |
| <b>NLP Insights</b>        | FR-8 | The system shall summarize forecast insights using natural language. |
| <b>Backup and Recovery</b> | FR-9 | The system shall automatically back up data daily.                   |

### 3.2 Non-Functional Requirements

| Category               | Requirement                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Performance</b>     | The system should handle at least <b>100 concurrent users</b> efficiently.                                                                                                |
| <b>Reliability</b>     | Forecasting accuracy should be at least <b>85%</b> based on test data.                                                                                                    |
| <b>Security</b>        | All user and transaction data should be <b>encrypted</b> in transit and at rest.                                                                                          |
| <b>Scalability</b>     | The system should support <b>increasing data volume</b> without major performance loss.                                                                                   |
| <b>Availability</b>    | The system should ensure <b>99% uptime</b> during operational hours.                                                                                                      |
| <b>Usability</b>       | The web interface should be <b>user-friendly</b> , responsive, and accessible on all devices.                                                                             |
| <b>Maintainability</b> | Codebase should be modular with clear documentation for future upgrades.                                                                                                  |
| <b>Backup</b>          | Daily automated backups of <b>MySQL</b> databases ( <b>0.5–0.7× DB size, 1–20 min</b> ) and <b>MongoDB</b> databases ( <b>0.3–0.5× DB size, 2–25 min</b> ) are performed. |